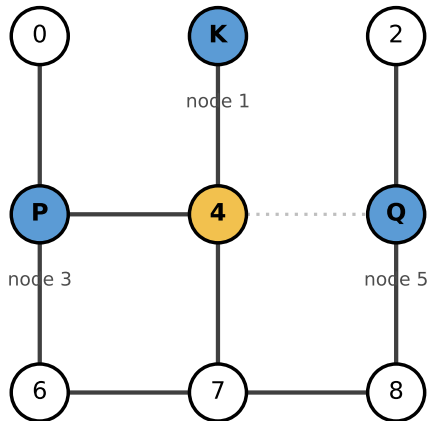


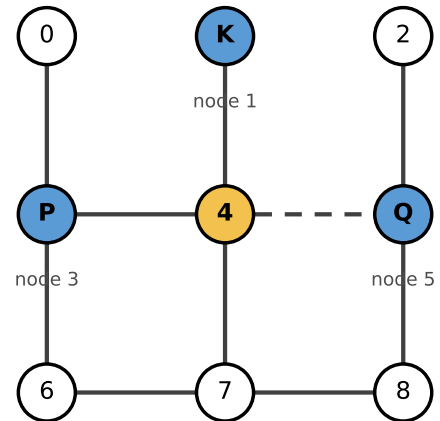
KILNWORKS — Retrofit Engineering Packet (Revision R3)

Section A — Plant aisle graph and docking bays

Closed-aisle designs (D0, D1, D2, D4)



Open-aisle designs (D3, D5)



Solid edges are permanent aisles. The dashed edge 4-5 exists only in the open-aisle retrofit (right). Filled blue nodes are docking bays (capacity 2 robots): P at node 3, Q at node 5, K at node 1. The amber node (4) is an interior node (capacity 1) shared by all traffic converging on K -- note that node 1 (K) has only one edge, to node 4, in BOTH layouts.

Section B — Six retrofit designs under evaluation

Design	F (fixtures)	G (power)	Aisle	Capital
D0	2	5	closed	0
D1	3	5	closed	7
D2	2	6	closed	9
D3	2	5	open	6
D4	3	6	closed	16
D5	3	6	open	22

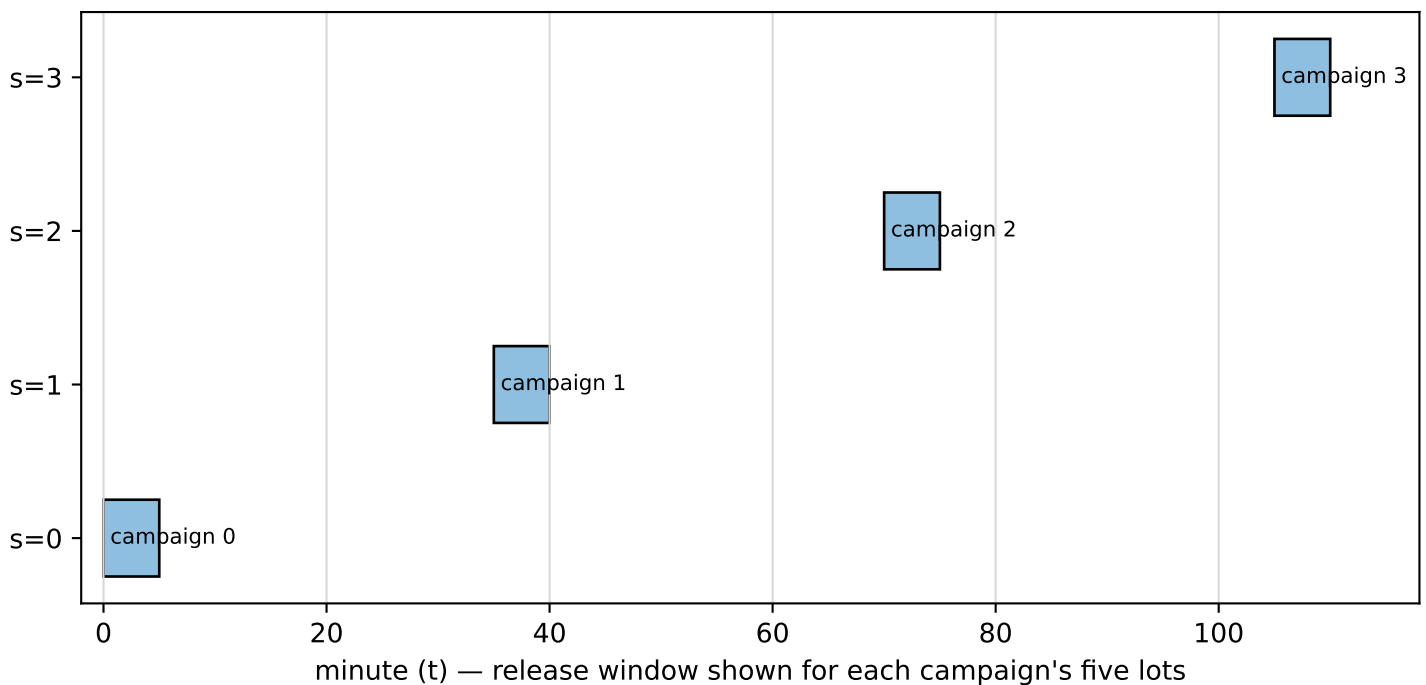
F = plant-wide fixture ceiling (a lot holds one fixture continuously from the start of its own preparation through curing). G = plant-wide aggregate power ceiling per minute. Capital is an investment figure used only for the final retrofit decision, never inside the simulation itself.

Section C — Campaign workload and cadence

Four campaigns ($s=0..3$), five lots each ($j=0..4$), 20 lots per design run end-to-end on one shared clock -- the four campaigns are NOT four independent resets. For lot (j,s):

```
family(j,s) = (j + s) mod 2
release local(j) = 2 * floor(j/2)
release abs(j,s) = 35*s + release local(j)      [cadence = 35 minutes]
Pbase(j,s) = 5 + ((j*j) + 2s) mod 4             (duration if processed on P)
Qbase(j,s) = 4 + (3j + s) mod 5                 (duration if processed on Q)
```

Campaign release windows share one clock and are spaced by a fixed cadence -- a campaign's own completion tail routinely extends past the next campaign's release window (see the full trace requirement in the prompt).



Section D — Preparation, fixtures, and machine memory

Each lot is prepared exactly once, on P or Q, in one uninterrupted setup-plus-processing operation; the machine then frees and the prepared lot waits in an unlimited buffer at that machine's dock until a robot collects it.

Each machine remembers the family of the last lot it processed – this memory is never reset between campaigns. Setup = 0 if this is the machine's first job ever, or the new job's family matches the remembered family; otherwise setup = 2.

Assignment policy each minute (P evaluated before Q): a machine that is busy is skipped. Otherwise, from lots that are released and not yet assigned AND for which a fixture is currently available (held fixtures < F), pick the one minimizing (setup + processing duration on this machine); tie-break by lowest global index (global index = $5s + j$). If a machine's pool is empty, or it cannot currently afford its power draw, it claims nothing this minute and does NOT remove anything from the other machine's pool. A fixture becomes held only at an actually-started preparation, releasing only when that lot's curing ends.

Section E — Robot dispatch and the node-4 disruption

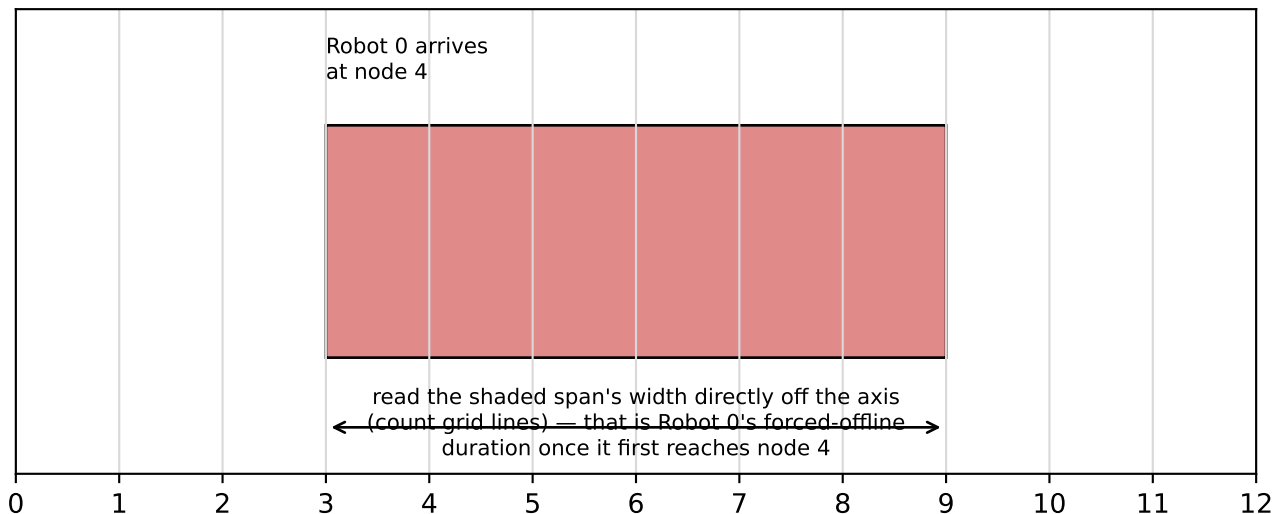
Robot 0 starts empty at node 3 (P); Robot 1 starts empty at node 5 (Q); once, at $t=0$ for the whole run. Each minute each robot takes exactly one action: wait, move one adjacent edge, pickup, or unload. Cargo capacity is 1. An action chosen at minute t becomes visible starting at minute $t+1$. Each edge carries at most one robot traversal per minute (no swap, no shared direction).

Robot 0's action for the minute is fully decided – including whether the plant's power budget actually admits it – before Robot 1's action is even computed. Priority order for each robot, first applicable wins:

1. inside its own forced-offline window -> no action at all;
2. carrying cargo and at K -> unload;
3. empty at a node with a waiting prepared lot -> pick up the earliest-prepared one (tie-break lowest global index);
4. carrying cargo, not at K -> move one step on the shortest path to K (tie-break lowest next-hop id), or wait if blocked/refused by power;
5. empty, no lot here -> move one step toward the NEAREST node holding a waiting prepared lot (tie-break lowest target-node id, then lowest next-hop id);
6. empty, no lot anywhere in the plant -> move one step toward its own home dock (3 for Robot 0, 5 for Robot 1); wait if already home.

Node 1 (K) has only the edge 1-4 in both aisle layouts (Section A), so every delivery to K passes through node 4 – the single shared junction – in every design regardless of the aisle retrofit.

Illustrative timeline — forced-offline window at first node-4 arrival
(the arrival minute itself is design-dependent; only the shaded span's length is normative)



Section F — Oven batching, power admission, and tariff

The oven holds one batch at a time; membership is fixed the moment a batch starts and never changes. A batch cures for (4 + family) minutes. Batching is by family value alone – lots from different campaigns MAY be batched together.

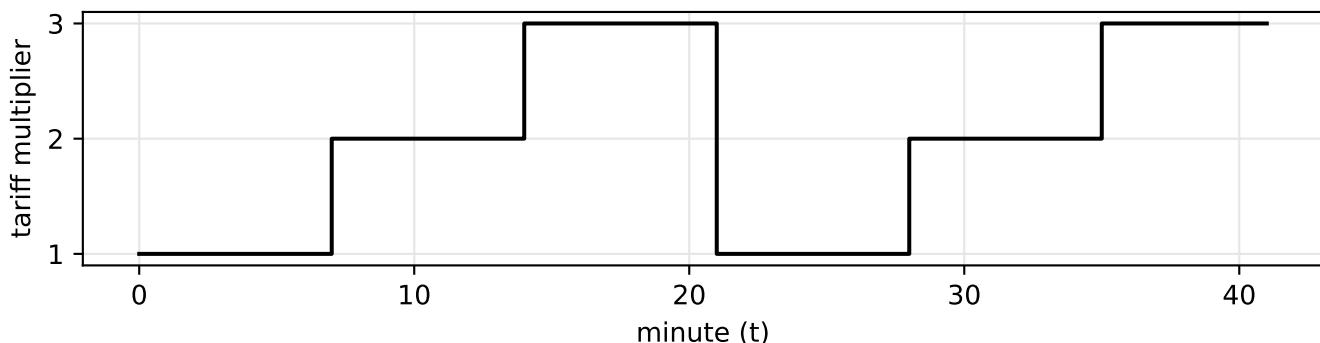
Bounded pairing timer (evaluated after both robots' actions, same minute eligible): a lot's "arrival minute" is the minute it is actually available per Section E's $t \rightarrow t+1$ visibility rule (one minute AFTER the unload action that delivers it, not the minute that action was decided). When the oven is idle with no pending anchor, the earliest-arrived waiting uncured lot (tie-break lowest global index) becomes the pending anchor, with deadline = its own arrival minute + 2. Every minute the anchor is pending and the oven is idle: if another arrived, uncured, same-family lot exists, start the batch now with the anchor plus the lowest-global-index such partner (size 2); else if the CURRENT minute is $\geq$ the deadline (deadline reached OR already passed; not strictly greater than), start with the anchor alone (size 1); otherwise wait. A power refusal does not forfeit the anchor or its deadline – it simply retries next minute.

Power admission is ONE ordered pass per minute, not two separate passes: machine P start, machine Q start, Robot 0's action, Robot 1's action, then a new oven start – each step drawing from whatever budget remains after the previous steps. Already-ongoing operations from an earlier minute are mandatory and keep their power. Draws: P=2, Q=3, oven=3, each robot move/pickup/unload=1, wait=0. Total draw must never exceed the design's G.

Section G — Investment selection

Selection key per design, minimized lexicographically (compare the first component; only break a tie with the next): (makespan, bill + 3*capital, capital, design ID). Unrestricted selection: the minimizing design among all six. Budget selection: the minimizing design among only those with capital ≤ 9 .

Tariff multiplier on that minute's total power draw: $\text{multiplier}(t) = 1 + (\text{floor}(t/7) \bmod 3)$. Bill = sum of power(t) x multiplier(t), $t = 0..makespan-1$.



Section H — Machine Q maintenance freeze

Track `Q_cumulative`: the sum of the `PROCESSING` portion only (never setup) of every job Q has completed since Q's last freeze ended (or since the start of the run, if no freeze has fired yet). At the boundary minute Q becomes free after completing a job, if adding that job's processing duration makes `Q_cumulative >= 10`, Q immediately becomes unable to start any new job AND `Q_cumulative` resets to 0 -- this is a recurring maintenance interval, not a one-time event: it fires again every time Q accumulates another 10 minutes of processing after resuming, with no cap on how many times it can fire in one design's run.

Unlike Section E's node-4 disruption, which explicitly states its freeze is inclusive of the triggering arrival minute, this rule states no such override -- so the `DEFAULT` timing convention applies (Section E: an action's effect becomes visible starting the minute AFTER it occurs) on EVERY trigger, not just the first: the triggering completion minute itself is unaffected, and each freeze covers the 13 minutes starting the minute after that trigger, through `trigger+13` inclusive. Q resumes normal assignment eligibility at `trigger+14`, and cumulative tracking resumes from 0 at that same point.

While frozen, Q simply never appears in the per-minute machine-assignment pass (P is unaffected and continues normally); a lot that would otherwise have gone to Q must wait for P or for Q's freeze to end, whichever the ordinary assignment-cost comparison and pool availability produce -- there is no special-case rerouting rule beyond applying the normal machine-assignment policy (Section D) to "Q currently ineligible."